

Executive Summary

Tracing Sparse Causal Features in an Open Language Model

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Purpose and importance

Large language models can produce correct answers without exposing the internal process that generated them. Mechanistic interpretability seeks to identify internal variables and computational pathways that causally support a model’s behaviour. Anthropic’s *On the Biology of a Large Language Model* presented an influential method that decomposes neural activations into sparse features, connects those features in attribution graphs, and tests graph-derived hypotheses by intervening on the model. Independent reproduction is important because an intuitively labelled graph is not necessarily a faithful account of the original computation.

This project tested whether the same scientific workflow could be transferred to Qwen3-4B-Instruct-2507, an open-weight model with four billion parameters. Two behaviours were studied: addition, focusing on propagation of a carry between decimal columns, and factual recall, represented by capital-city and SI unit questions. The objective was not merely to generate diagrams, but to determine whether small groups of learned features exerted the predicted and concept-specific causal effects.

Approach

An independent, public pipeline generated synthetic prompts, recorded multilayer perceptron activations at seven transformer layers, trained a sparse autoencoder for each layer, constructed contrastive attribution graphs, and performed feature inhibition and activation swap-in experiments. Complete autoencoder codes and raw model activations were patched as positive controls. Clean-baseline checks ensured that each evaluated prompt expressed the expected behaviour.

The investigation was deliberately sequential. Initial feature interventions were weak, so reconstruction and sparsity were measured directly. The first autoencoders explained 90.7–95.9% of held-out activation variance, but used 2,167–3,545 of 8,192 latent variables per example. The problem was therefore not simply inaccurate reconstruction: the codes were too dense to support a compact feature-level account. A fixed comparison of TopK autoencoders then tested whether direct control of the number of active variables improved causal specificity. Candidate models were selected from reconstruction diagnostics before their intervention outcomes were examined.

Key findings

Broad activation patches demonstrated that the selected model components contained causally relevant information. In addition, replacing target MLP states with those from a matched source could transfer the source answer digit. However, a source with a different answer digit transferred that digit as well, so the broad intervention moved general answer state rather than isolating carry. TopK retraining made sparse arithmetic interventions larger, but graph-derived and predeclared ten-feature panels failed matched no-carry specificity tests. A final analysis controlled explicitly for the output digit and ranked all 57,344 SAE latents on separate discovery cases. Its ten-feature primary still failed, but a secondary 20-feature effect was frozen and tested once on 32 intervention-untouched cases. Inhibition reduced the carry-target logit gap by 0.059 logits while the no-carry control increased by 0.027, giving a paired effect of -0.086 (bootstrap 95% interval -0.137 to -0.039). This supports a distributed carry-associated panel, although no answer flipped and the same SAE basis was reused. Capital-city swaps remained weak after low-confidence examples were replaced with high-confidence prompts.

The strongest positive result occurred for SI units. The selected TopK model retained 93.6% mean held-out variance while using approximately 100 active latents, 35.5 times fewer than the original units autoencoder. Features from one force attribution graph were ranked on eight discovery systems. The resulting ten-feature panel was frozen and evaluated on 16 different physical systems. Swapping force-source values into energy prompts increased the newtons-versus-joules logit gap by 1.234 logits (bootstrap 95% interval 1.148–1.332). A matched mass-source control changed the gap by -0.090 (-0.129 to -0.051), giving a paired specificity effect of 1.324 logits (1.223–1.430). Every confirmation case moved more strongly under the force source than under the mass control. Eight of the ten latents were located at layer 28, and the panel reproduced 97.5% of the force-source shift obtained by swapping all 73 positive graph features.

The model nevertheless continued to predict joules in every confirmation case. The result is therefore evidence for a compact, force-associated causal panel, not a complete factual-recall circuit, a set of uniquely necessary variables, or a transferred answer.

Implications and next steps

The project provides a partial reproduction with a clear inferential boundary. Sparse SAE features can exert compact, concept-selective causal effects in a lightweight open-model setting. The force panel met a predeclared criterion, while the smaller arithmetic effect emerged only after an initial primary failure and therefore required a separate frozen replication. Reconstruction accuracy, decodability and directional logit movement alone were insufficient; both positive claims depend on matched controls and held-out cases.

The highest-priority next step is replication of both selection procedures across independently trained autoencoder seeds, using larger frozen confirmation pools and more random feature panels. Arithmetic autoencoders should be trained on the same teacher-forced decimal positions used in interventions, with carry status and answer digit varied independently. At greater computational scale, per-layer or cross-layer transcoders and explicit error nodes would provide a closer architectural comparison with Anthropic’s method.

All principal code, configurations, notebooks and output formats are documented in the public repository at https://github.com/evey-dev/test_run. The reproducibility pack allows the positive units result, the controlled negative results, and the methodological diagnostics to be audited independently.